



GOOGLE PLAY STORE ANALYTICS DASHBOARD

Business Intelligence Project Report

Project Focus

App performance • Categories • Installs • Ratings • Reviews • Pricing • Sentiment

1. Executive Summary

This project presents an interactive analytics dashboard for exploring Google Play Store app data. The dashboard converts app-level information into business-oriented views covering app volume, install performance, category distribution, ratings, reviews, pricing type, and review sentiment.

The supplied dashboard contains multiple analytical pages. The first page provides an overall market snapshot, while the subsequent pages drill into app performance, free-versus-paid applications, category-level review behavior, and sentiment. Together, the views are designed to help a business user identify high-performing categories, popular applications, engagement patterns, and differences between free and paid apps.

2. Project Objectives

- Understand the overall distribution of applications across categories.
- Measure app popularity using installs and review activity.
- Compare category-level performance using app count, installs, ratings, and reviews.
- Identify highly installed applications and understand their contribution to overall performance.
- Compare free and paid applications.
- Analyze review volume and sentiment patterns across categories.
- Create an interactive dashboard that supports filtering and visual exploration.
- Translate raw app data into concise KPIs and business insights.

3. Dataset & Scope

The dashboard appears to be based on a Google Play Store application dataset. The screenshots show approximately 9K apps, with dashboard-level figures of about 65bn total installs and 37K total reviews displayed in the KPI area. Individual charts use category, app, installs, ratings, review counts, pricing type, and sentiment-related fields.

Area	Dashboard Measure	Purpose
App Inventory	Total Apps / Rated Apps	Measure application coverage
Popularity	Total Installs	Measure adoption
Engagement	Reviews / Ratings	Measure user interaction
Category	App Count / Installs / Reviews	Compare app segments
Commercial Model	Free vs Paid	Compare pricing types
Sentiment	Positive / Negative / Neutral	Summarize review tone

Note: The exact source-file row count, cleaning rules, sentiment model, and data-refresh date are not visible in the screenshots. Those items should be added if the original dataset/documentation is available.

4. Dashboard Structure

Page 1 — Executive Overview

KPI cards for total apps, rated apps, total installs, and total reviews; category-wise app count; category distribution by year; and category-level install counts.

Page 2 — App Performance & Relationship Analysis

Top applications by installs, category-level rating/review comparison, and a bubble-style relationship view between rating count and review volume.

Page 3 — Pricing & Category Analysis

Free versus paid application distribution, category-wise app counts, and review volume by application type.

Page 4 — Reviews & Sentiment

Category-wise review totals, a detailed category table containing app counts, installs and review measures, and a sentiment distribution across positive, negative, and neutral reviews.

5. Dashboard Screenshots

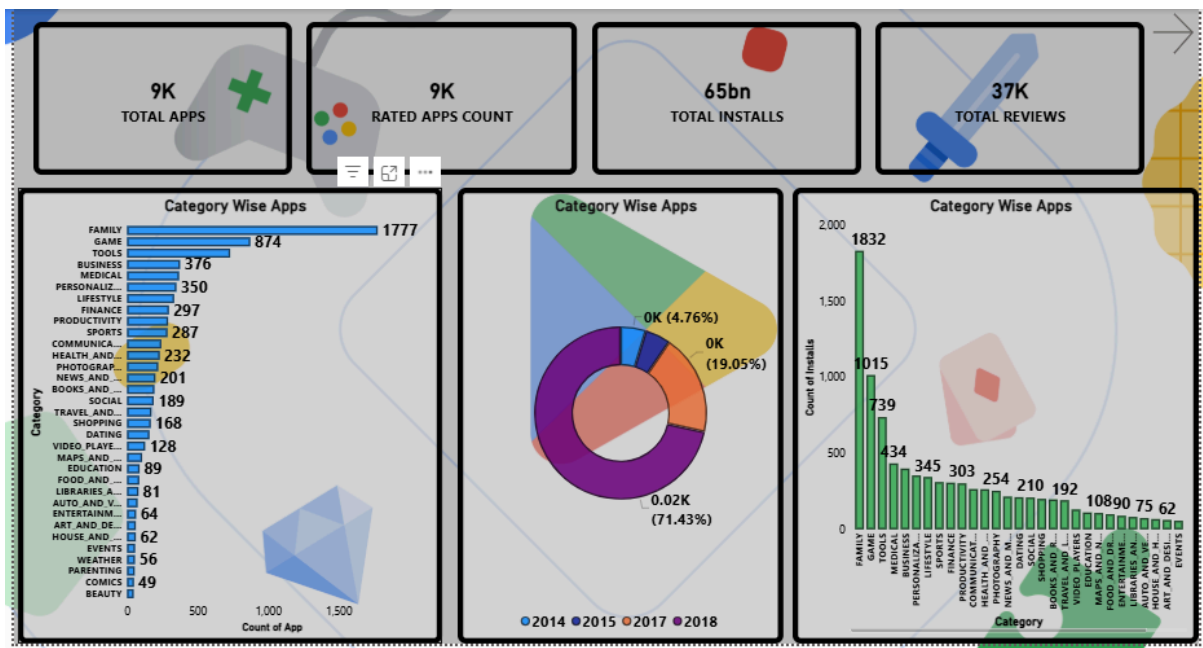


Figure 1. Executive overview—KPI and category analysis.

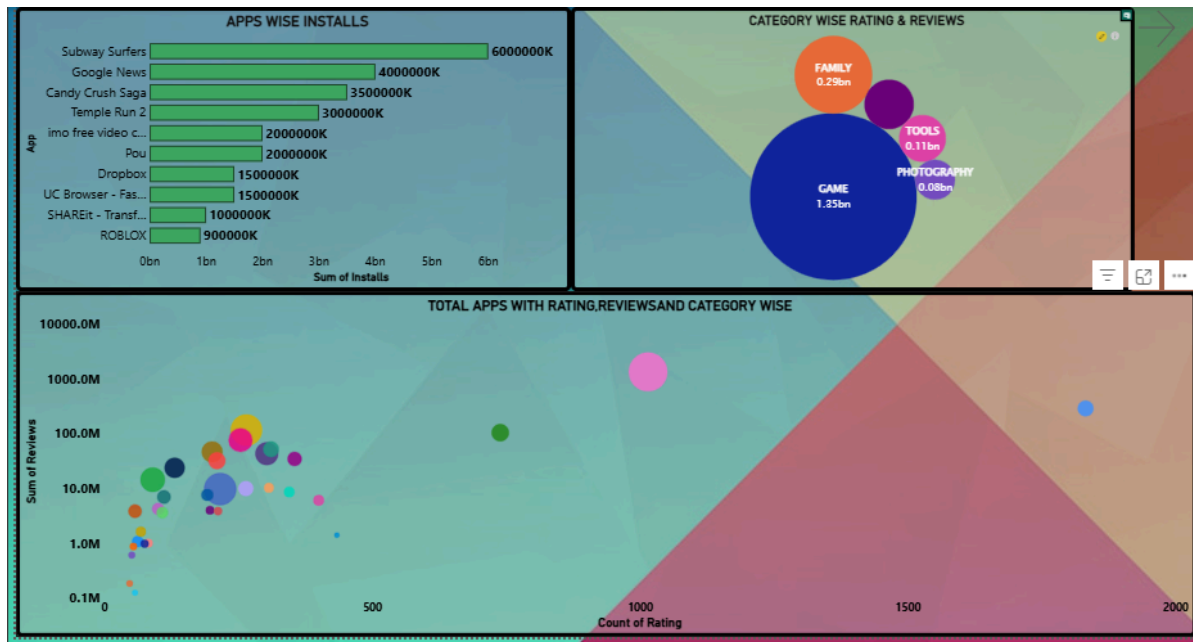


Figure 2. App installs, category rating/review analysis, and relationship view.



Figure 3. Free vs paid, category, and type-wise review analysis.



Figure 4. Category reviews, detailed category table, and sentiment analysis.

6. Key Analytical Observations

- The dashboard shows a large application ecosystem, with the main KPI cards displaying approximately 9K total apps.
- Install activity is highly concentrated in a relatively small number of applications. The top-app chart shows Subway Surfers, Google News, Candy Crush Saga, Temple Run 2 and other applications with very large install totals.
- The category charts indicate that Family has the largest app count in the displayed category ranking, followed by Game and other categories.
- Game is the dominant category in the category-wise review visualization, with Family and Tools also showing substantial review volumes.
- The pricing page shows that free applications make up the overwhelming majority of the displayed app inventory, while paid apps represent a much smaller share.
- The type-wise review chart shows a very large difference in review volume between free and paid applications.
- The review dashboard separates sentiment into positive, negative, and neutral groups, allowing user feedback tone to be considered alongside review volume.
- The category detail table is useful for moving from high-level visuals to category-level comparison of app counts, installs, and review measures.

7. Business Questions Answered

Business Question	Dashboard Evidence
Which categories contain the most applications?	Category-wise app count ranking.
Which applications have the highest install volumes?	Top Apps by Installs chart.
Which categories generate high review activity?	Category-wise review analysis.
How do free and paid apps differ?	Free vs Paid and type-wise Reviews visuals.
Is app popularity concentrated among a few applications?	Top-install chart and rating/review relationship view.
What is the distribution of review sentiment?	Sentiment-wise reviews, Donut chart.

8. Technical Implementation

The screenshots show a Power BI-style dashboard environment. The report therefore treats Power BI as the visualization and dashboarding layer. A typical implementation for this project consists of data preparation, data modeling, DAX measures, visual design, and interactive filtering.

- **Data Preparation:** standardize category, app, pricing, install, rating, and review fields; handle missing values and inconsistent formats.
- **Data Modeling:** Create a clean analytical table or a star-schema model where appropriate.
- **Measures:** build KPIs for app count, installs, ratings, reviews, and category/type comparisons.
- **Visualization:** use KPI cards, bar charts, donut charts, bubble/scatter visuals, and matrix/table views.
- **Interactivity:** use slicers, cross-filtering, drill-through, or page navigation where supported.
- **Presentation:** maintain consistent titles, units, number formatting, and category labels.

9. Core Measures

- Total Apps = DISTINCTCOUNT(App[App])
- Total Reviews = SUM(App[Reviews])
- Total Installs = SUM(App[Installs])
- Average Rating = AVERAGE(App[Rating])
- Free Apps = CALCULATE([Total Apps], App[Type] = "Free")
- Paid Apps = CALCULATE([Total Apps], App[Type] = "Paid")
- Positive Reviews = CALCULATE([Total Reviews], App[Sentiment] = "Positive")
- Negative Reviews = CALCULATE([Total Reviews], App[Sentiment] = "Negative")
- Neutral Reviews = CALCULATE([Total Reviews], App[Sentiment] = "Neutral")

These are implementation-based on the fields used in the dashboard; exact DAX syntaxes are applied during data cleaning and data modelling.

10. Data Quality & Validation Recommendations

- Standardize install values before aggregation (for example, convert text such as '1,000,000+' into a numeric convention).
- Validate whether review counts and install counts are stored as numeric values or formatted text.
- Check duplicate app names and decide whether duplicate records represent different versions/records or data duplication.
- Handle missing ratings explicitly and distinguish 'no rating' from a numerical zero.
- Validate category labels and merge accidental duplicates caused by spelling or formatting differences.
- Document the methodology used to derive 'Positive', 'Negative', and 'Neutral sentiment'.
- Reconcile KPI totals against source data after every major cleaning step.

11. Dashboard Enhancement Opportunities

- Add slicers for Category, Type, Content Rating, Android Version, and Last Updated date if those fields exist in the source data.
- Add a time-series view for installs, ratings, or reviews if historical/date fields support meaningful trend analysis.
- Add tooltip pages with app-level details such as rating, reviews, installs, price, and category.
- Use dynamic titles so users can immediately see which filters are active.
- Add a Top-N selector so users can switch between Top 5, Top 10, Top 20, etc.
- Add an install-to-review engagement ratio to compare adoption with visible user feedback.
- Improve accessibility by increasing label readability, reducing background decoration, and keeping chart scales consistent.

12. Conclusion

The Google Play Store Analytics Dashboard provides a multi-page analytical view of application performance. It combines executive KPIs with category, application, pricing, review, and sentiment analysis. The project demonstrates how raw app-store data can be transformed into an interactive business intelligence product that supports market segmentation, popularity analysis, engagement analysis, and review interpretation.

For a portfolio or academic submission, the project can be strengthened further by documenting the source dataset, data-cleaning workflow, data model, DAX measures, sentiment methodology, and validation checks.

13. Appendix — Dashboard-to-Insight Mapping

Dashboard Visual	Primary Metric	Analytical Use
KPI Cards	Apps / Installs / Reviews	Executive snapshot
Category-Wise Apps	Count of Apps	Category size
Apps Wise Installs	Sum of Installs	Application popularity
Category-Wise Rating & Reviews	Ratings / Reviews	Category engagement
Free vs Paid	App count	Pricing mix
Type-Wise Reviews	Review volume	Engagement by pricing type
Category-Wise Reviews	Review volume	Category engagement
Sentiment-Wise Reviews	Sentiment count	Review tone.